Himanshu Kumar

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Bachelor of Technology
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GitHub | LinkedIn
LeetCode | CodeChef

Education

•Bachelor of Technology in Information Technology *2022-26*
Rajiv Gandhi Institute of Petroleum Technology, Hoskote, Bengaluru CGPA: 8.36

Research & Publication | [Link]

•A DEC Method for Location-Specific Driving Safety Profiling using Trajectory Data *2025*
Manuscript accepted *Nature Journal*

- Contributed as **Author** (Kushwaha, A. K., **Kumar, H.**, & Kumar, B. A.) by implementing algorithmic refinements, re-running experiments, and addressing reviewer comments to significantly improve manuscript quality.

Work Experience

•Indian Institute of Technology Patna | [Certificate] *May - Jul 2025*
Data Science Intern Patna, India

- Extracted and processed sensor data (LiDAR, accelerometers) from 2,200+ drivers, building a driver behavior model focused on speed, acceleration, and direction-based features.
- Enhanced driver behavior model through implementation of scaling and oversampling techniques; selected and trained K-Means, Mean Shift, DBSCAN and DEC clustering models with optimized parameters.
- Improved model performance using GridSearchCV(Hyperparameter tuning), and clustering metrics (Silhouette, Elbow, CHI, DBI); successfully predicted driver traits namely Aggressive, Speedy and Calm

Projects

•AI Resume Analyzer | Spring Boot, Java, HuggingFace API|
Developed an AI-powered application that analyzes resumes with job descriptions, and generates personalized interview preparation.

- Designed and implemented a Spring Boot–based application following layered architecture (Controller–Service–Repository) to process resume and job description inputs.
- Model & Tools Used:Integrated HuggingFace NLP APIs to perform semantic analysis of resumes and implemented a resume–job description matching engine to compute compatibility scores based on skill and contextual similarity.
- Secure login with MySQL data and generated personalized interview questions based on resume–job description.

•Weather Information Web Application (Java J2EE, Servlets, JSP, MVC, REST API) |
Developed a dynamic web application using traditional Java J2EE architecture to display real-time weather data ,city wise

- Implemented the application using Servlets and JSP, following the Model–View–Controller (MVC) design pattern.
- Integrated the OpenWeatherMap REST API and implemented HTTP request handling and response processing using Java Servlets to retrieve and display real-time weather data.
- Implemented dynamic web pages using JSP for server-side rendering, parsed JSON responses from external APIs to extract weather data, and designed a responsive user interface using HTML, CSS, and JavaScript to display real-time weather information.

Technical Skills and Interests

Languages/Frameworks:C++Java,HTML5, CSS3, JS, ReactJS (**Frontend**)
Database/Server: MongoDB, MySQL, NodeJS, ExpressJS (**Backend**)
Libraries/Tools: pandas, NumPy, Matplotlib, Github, Jupyter
Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOPs, DBMS, Software Engineering.
Areas of Interest: Web Design and Development, AI/ML.

Achievements

- Springer Nature–accepted research paper (2025)
- Awarded a prestigious sponsorship by HPCL to attend **India Energy Week (IEW) 2026.**
- Finalist at **IIM Bangalore Ideathon 2.0** and Pre-finalist in the **IIT BHU DPBH [Certificate]** *2024*
- **JEE Advanced** Among Top 2 percent of the country *2022*
- **LeetCode and CodeChef** Solved **300**+ DSA Problems *2023-25*

Positions of Responsibility

- **Training and Placement Coordinator** Organized and led recruitment activities for the CSE department.
- **Teaching Assistant**, mentored 80+ students in learning Data Structures and Algorithms (DSA).